Normal Distribution

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Outline

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3.

Introduction to Normal Distribution

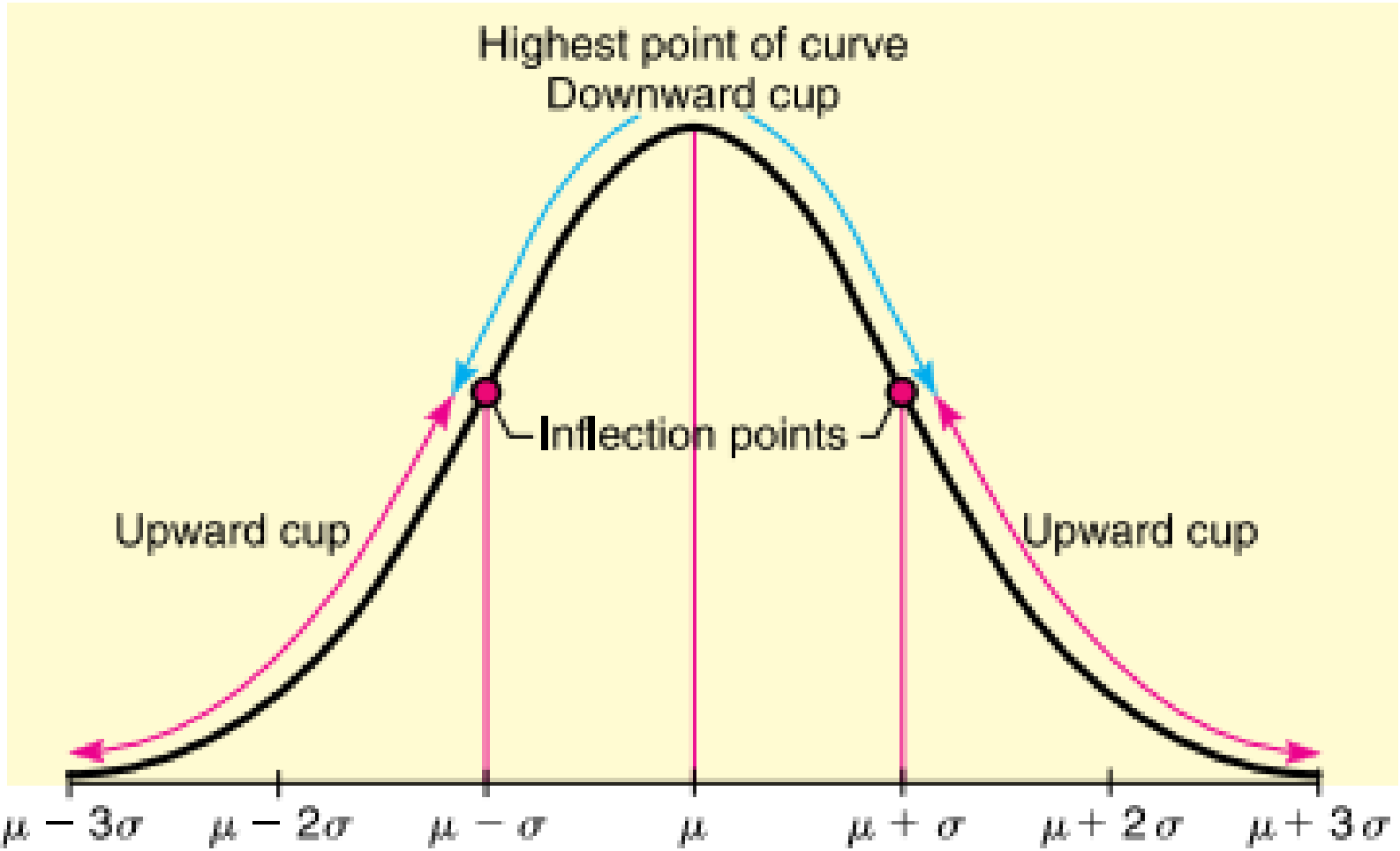
- We will study the most important type of probability distribution in all of mathematical statistics: the normal distribution. Why is the normal distribution so important?
- Two of the reasons are that **it applies to a wide variety of situations and that other distributions tend to become normal under certain conditions.**
- There is an old saying: All roads lead to Rome. in statistics, we could recast this saying: All probability distributions average out to be normal distributions (as the sample size increases).

Preview questions

- What are some characteristics of a normal distribution? What does the empirical rule tell you about data spread around the mean?
- How can this information be used in quality control? Can you compare apples and oranges, or maybe elephants and butterflies? In most cases, the answer is no-unless you first standardize your measurements. What are a standard normal distribution and a standard z score?
- How do you convert any normal distribution to a standard normal distribution? How do you find probabilities of "standardized events"?

Graphs of Normal Probability Distributions

- One of the most important examples of a continuous probability distribution is the normal distribution.
- This distribution was studied by the French mathematician **Abraham de Moivre (1667–1754)** and later by the German mathematician **Carl Friedrich Gauss (1777–1855)**, whose work is so important that the normal distribution is sometimes called **Gaussian**.
- Applications of a normal probability distribution are so numerous that some mathematicians refer to it as "a veritable Boy Scout knife of statistics."
- Normal distribution in is defined in terms of μ and σ (or σ^2) , the mean and standard deviation (or variance) of the population distribution.

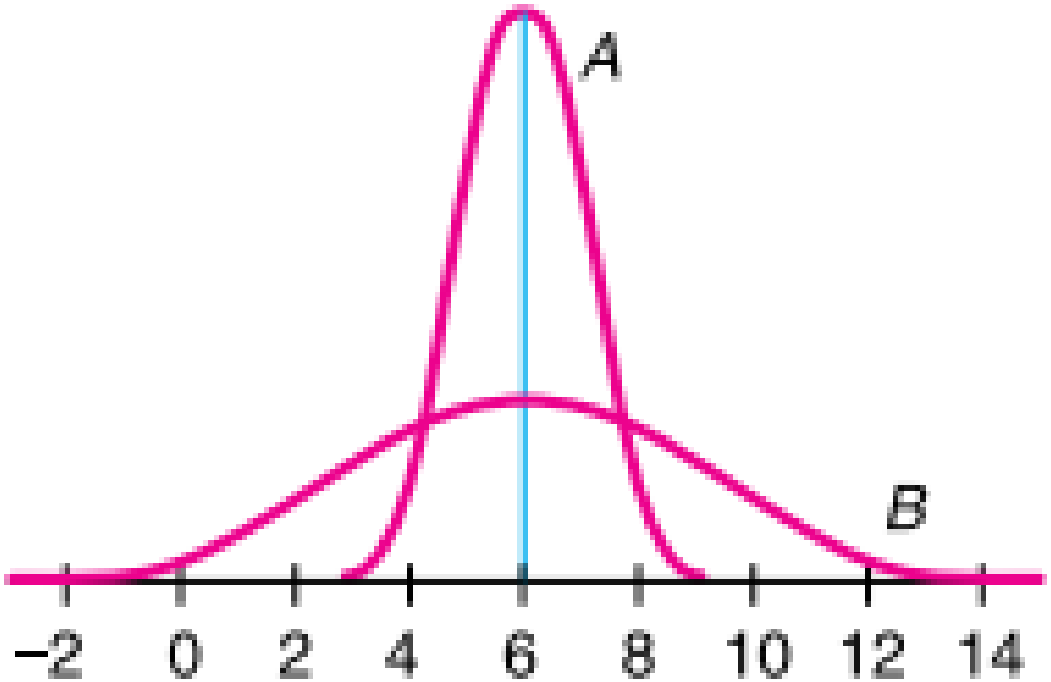

Highest point of curve
Downward cup
Inflection points
Upward cup
Upward cup
$\mu - 3\sigma$
$\mu - 2\sigma$
$\mu - \sigma$
μ
$\mu + \sigma$
$\mu + 2\sigma$
$\mu + 3\sigma$

- The graph of a normal distribution is called **a normal curve.** It possesses a shape very much like the cross section of a pile of dry sand.
- Because of its shape, blacksmiths would sometimes use a pile of dry sand in the construction of a mold for a bell. Thus the normal curve is also called **a bell-shaped curve.**
- A general normal curve is smooth and symmetric about the vertical line extending upward from the mean μ. Notice that the highest point of the curve occurs over μ.
- We also see that the curve tends to level out and approach the horizontal (x axis) like a glider making a landing. However, in mathematical theory, such a glider would never quite finish its landing because a normal curve never touches the horizontal axis.

- **The parameter s controls the spread of the curve. The curve is quite close to the horizontal axis at $\mu + 3\sigma$ and $\mu - 3\sigma$.**
- **Thus, if the standard deviation σ is large, the curve will be more spread out; if it is small, the curve will be more peaked.**
- The exact places where the transition between the upward and downward cupping occur are above the points $\mu + \sigma$ and $\mu - \sigma$.
- *In the terminology of calculus, transition points such as these are called* **inflection points.**

Important properties of a normal curve

1. The curve is bell-shaped, with the highest point over the mean μ.
2. The curve is symmetric about a vertical line through μ.
3. The curve approaches the horizontal axis but never touches or crosses it.
4. The inflection (transition) points between cupping upward and downward occur above $\mu + \sigma$ and $\mu - \sigma$.
5. The area under the entire curve is 1.


A
B
−2
0
2
4
6
8
10
12
14

- The area beneath the curve and above the axis is exactly 1. As such, the normal distribution curve is an example of **a density curve.**
- The formula used to generate the shape of the normal distribution curve is called the **normal density function.**
- f x is a normal random variable with mean μ and standard deviation σ, the formula for the normal density function is

$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{1}$$

CHEBYSHEV'S THEOREM

- For any set of data (either population or sample) and for any constant $k > 1$, the proportion of the data that must lie within k standard deviations on either side of the mean is at least

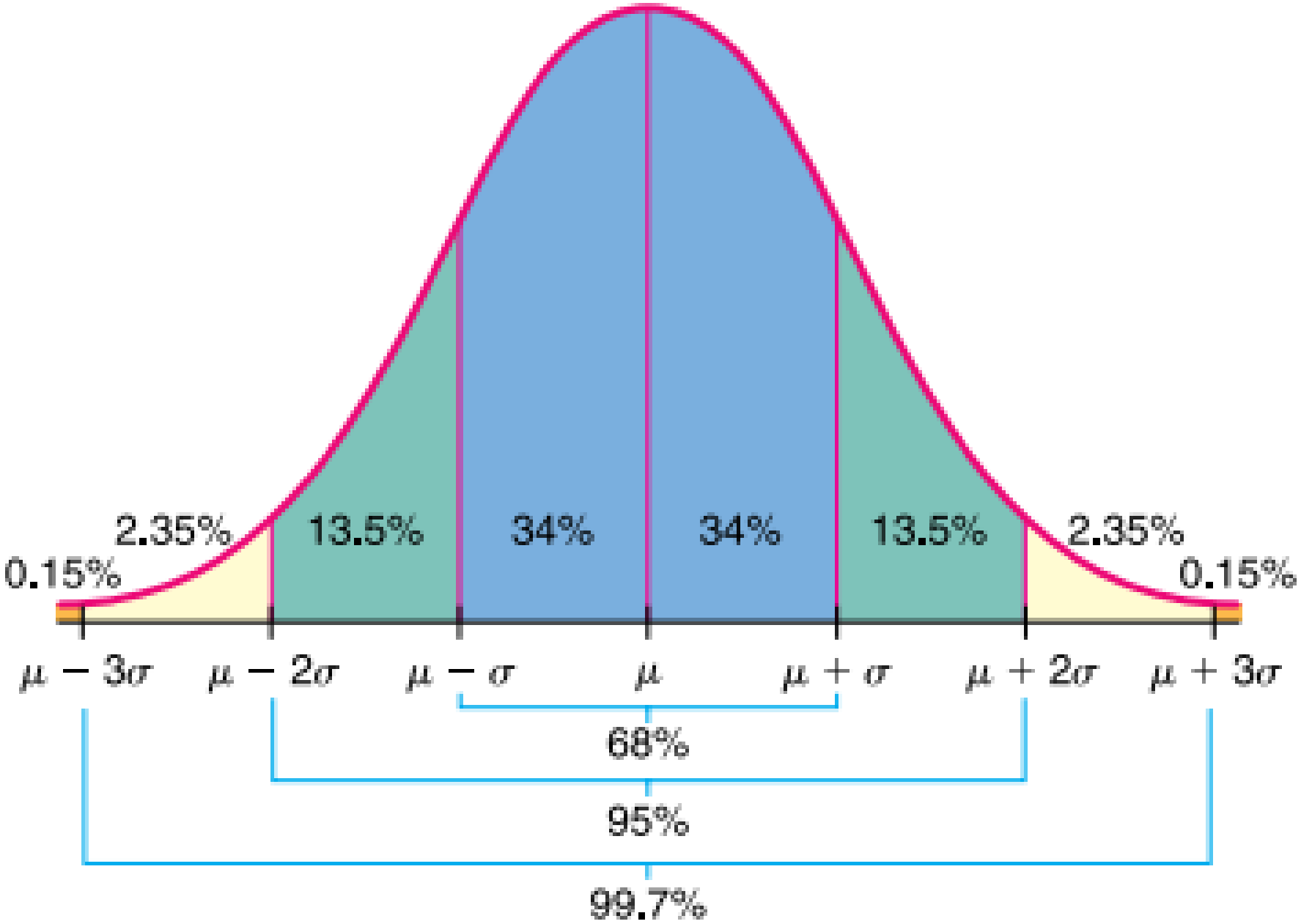
$$\boxed{1 - \frac{1}{k^2}} \tag{2}$$

- Put another way: For data with mean μ and standard deviation σ, at least $\boxed{1 - \frac{1}{k^2}}$ (fractional part) of data must fall between $\mu - k\sigma$ and $\mu + k\sigma$.

Emperical rule

- For a distribution that is symmetric and bell-shaped (in particular, for a normal distribution):
  1. Approximately 68% of the data values will lie within 1 standard deviation on each side of the mean.
  2. Approximately 95% of the data values will lie within 2 standard deviations on each side of the mean.
  3. Approximately 99.7% (or almost all) of the data values will lie within 3 standard deviations on each side of the mean.
- **NB.: Notice that the empirical rule is a stronger statement than Chebyshev's theorem in that it gives definite percentages, not just lower limits. Of course, the empirical rule applies only to normal or symmetric, bell-shaped distributions, whereas Chebyshev's theorem applies to all distributions.**

Area Under a Normal Curve



- Economic Internet Service has user connection times that are normally distributed with a mean $\mu = 5$ seconds and a standard deviation $\sigma = 1.5$ seconds. What is the probability that a connection selected at random takes from 5 to 6.5 seconds?

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SOLUTION: The probability that the connection time will be between 5 and 6.5 seconds is equal to the percentage of total area under the normal curve that is shaded in Figure 6-4. Since $\mu = 5$ and $\mu + \sigma = 5 + 1.5 = 6.5$, we see that the shaded area is the area between μ and $\mu + \sigma$. By the empirical rule, the area from μ to $\mu + \sigma$ is 34% of the total area. This tell us that the probability a connection time for Economic Internet will be between 5 and 6.5 seconds is about 0.34.

control charts

- If we are examining data over a period of equally spaced time intervals or in some sequential order, then control charts are especially useful.
- Business managers and people in charge of production processes are aware that there exists an inherent amount of variability in any sequential set of data.
- The sugar content of bottled drinks taken sequentially off a production line, the extent of clerical errors in a bank from day to day, advertising expenses from month to month, or even the number of new customers from year to year are examples of sequential data. There is a certain amount of variability in each.
- A random variable X is said to be in *statistical control* if it can be described by the same probability distribution when it is observed at successive points in time.

control charts

- Control charts combine graphic and numerical descriptions of data with probability distributions.
- Control charts were invented in the 1920s by Walter Shewhart at Bell Telephone Laboratories.

How to make a control chart for the random variable X

A control chart for a random variable x is a plot of observed x values in time sequence order.

1. Find the mean μ and standard deviation σ of the x distribution by
   1. using past data from a period during which the process was "in control" or
   2. using specified "target" values for μ and σ.
2. Create a graph in which the vertical axis represents x values and the horizontal axis represents time.
3. Draw a horizontal line at height μ and horizontal, dashed control-limit lines at $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$.
4. Plot the variable x on the graph in time sequence order. Use line segments to connect the points in time sequence order.

- Susan Tamara is director of personnel at the Antlers Lodge in Denali National Park, Alaska. Every summer Ms. Tamara hires many part-time employees from all over the United States. Most are college students seeking summer employment. One of the biggest activities for the lodge staff is that of "making up" the rooms each day. Although the rooms are supposed to be ready by 3:30 p.m., there are always some rooms not made up by this time because of high personnel turnover. Every 15 days Ms. Tamara has a general staff meeting at which she shows a control chart of the number of rooms not made up by 3:30 p.m. each day. From extensive experience, Ms. Tamara is aware that the distribution of rooms not made up by 3:30 p.m. is approximately normal, with mean $\mu = 19.3$ rooms and standard deviation $\sigma = 4.7$ rooms. This distribution of x values is acceptable to the top administration of Antlers Lodge. For the past 15 days, the housekeeping unit has reported the number of rooms not ready by 3:30 p.m.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
x	11	20	25	23	16	19	8	25	17	20	23	29	18	14	10

Figure: number of rooms x not made up by 3:30 p.m

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
x	11	20	25	23	16	19	8	25	17	20	23	29	18	14	10

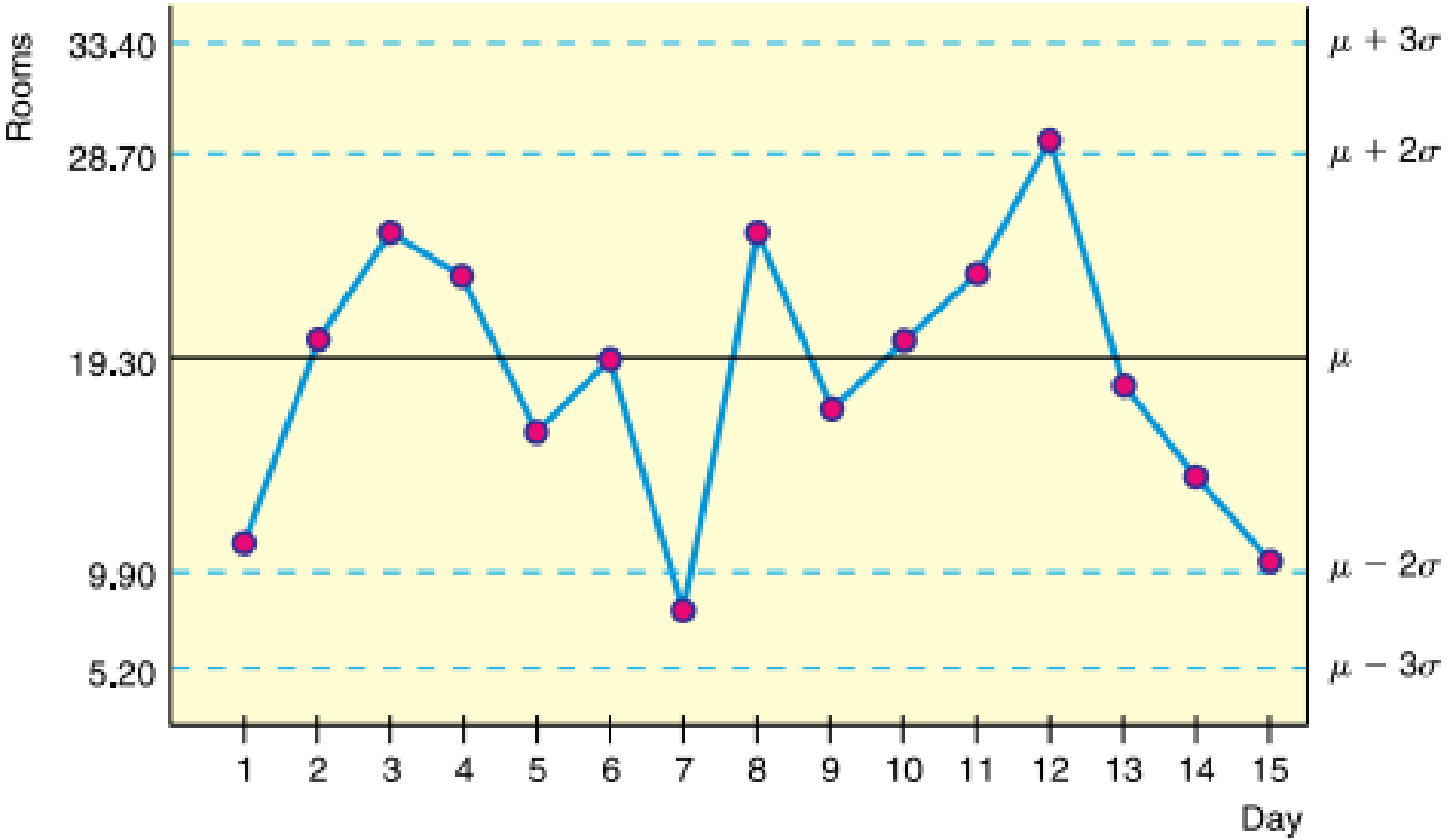
Figure: number of rooms x not made up by 3:30 p.m

SOLUTION: A control chart for a variable x is a plot of the observed x values (vertical scale) in time sequence order (the horizontal scale represents time). Place horizontal lines at

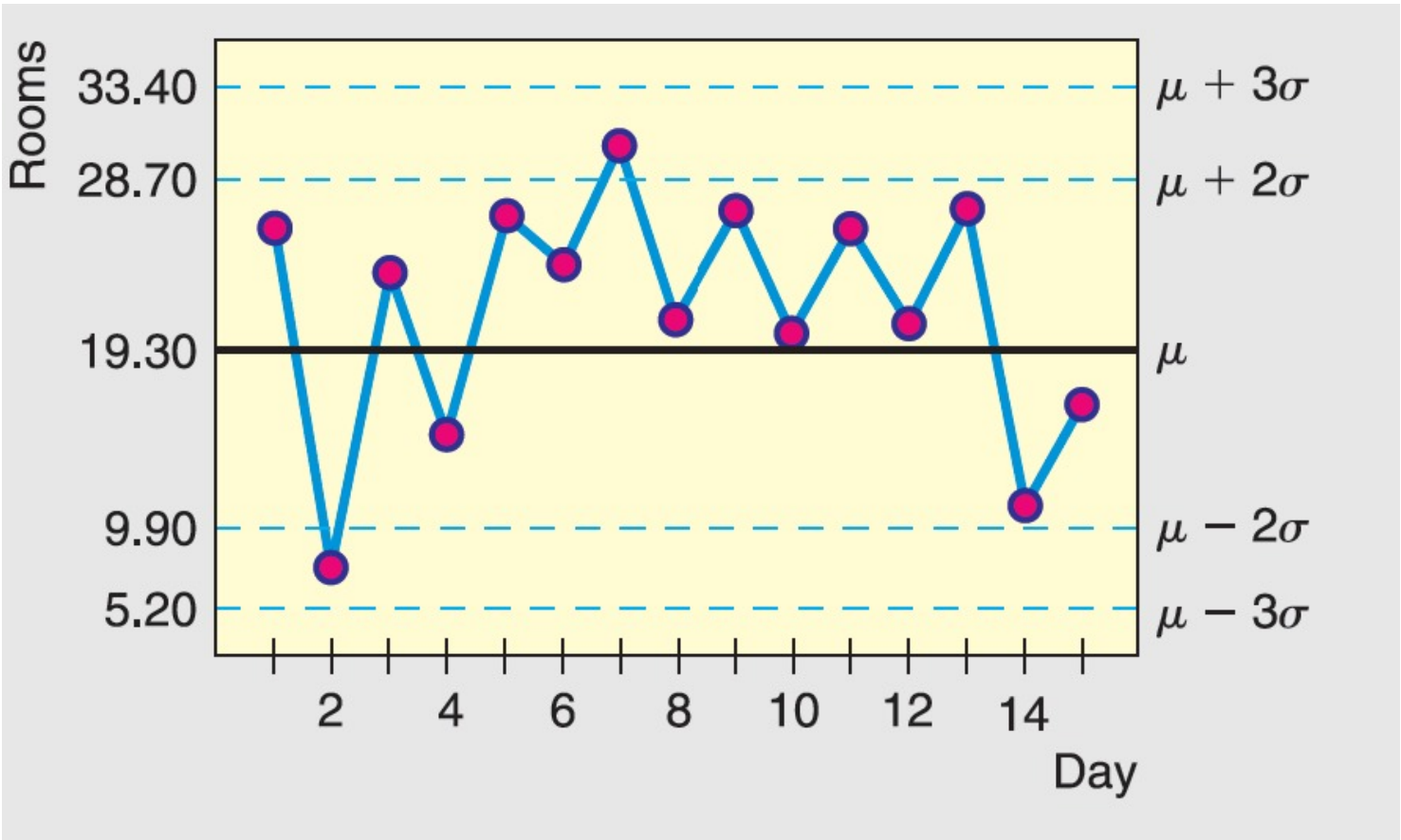
the mean $\mu = 19.3$

the control limits $\mu \pm 2\sigma = 19.3 \pm 2(4.7)$, or 9.90 and 28.70

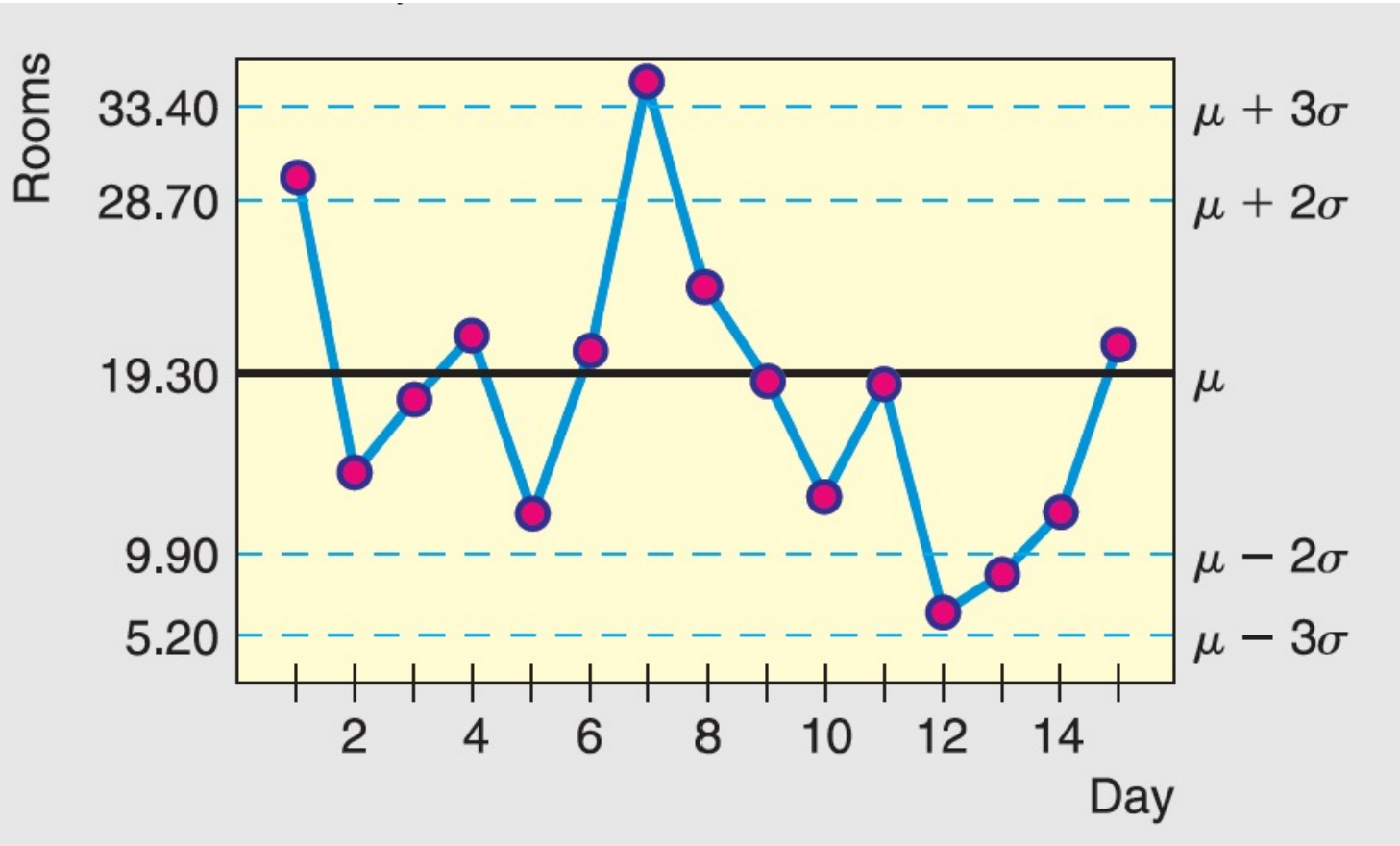
the control limits $\mu \pm 3\sigma = 19.3 \pm 3(4.7)$, or 5.20 and 33.40


Rooms
33.40
28.70
19.30
9.90
5.20
$\mu + 3\sigma$
$\mu + 2\sigma$
μ
$\mu - 2\sigma$
$\mu - 3\sigma$
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Day

Ms. Tamara of the Antlers Lodge examines the control chart for housekeeping. During the staff meetings, she makes recommendations about improving service or, if all is going well, she gives her staff a well-deserved "pat on the back." Look at the control chart created in Figure below to determine if the housekeeping process is out of control.



Days 5 to 13 are above $\mu = 19.3$. We have nine consecutive days on one side of the mean. This is a warning signal! It would appear that the mean m is slowly drifting up beyond the target value of 19.3. The chart indicates that housekeeping is "out of control." Ms. Tamara should take corrective measures at her staff meeting.



On day 7, we have a data value beyond $\mu + 3\sigma$(i.e., above 33.40). On days 11, 12, and 13, we have two of three data values beyond $\mu - 2\sigma$ (i.e., below 9.90). The occurrences during both of these periods are out-of-control warning signals. Ms. Tamara might ask her staff about both of these periods. There may be a lesson to be learned from day 7, when housekeeping apparently had a lot of trouble.

Standard Units and Areas Under the Standard Normal Distribution

- The **z value or z score** (also known as standard score) gives the number of standard deviations between the original measurement x and the mean μ of the x distribution.

$$\boxed{z = \frac{x - \mu}{\sigma}} \tag{3}$$

- When $x = \mu$,

$$z = \frac{x - \mu}{\sigma} = z = \frac{\mu - \mu}{\sigma} = 0 \tag{4}$$

- The mean of the original distribution is always zero, in standard units.
- This makes sense because the mean is zero standard variations from itself.
- An x value in the original distribution that is above the mean m has a corresponding z value that is positive. Again, this makes sense because a measurement above the mean would be a positive number of standard deviations from the mean. Likewise, an x value below the mean has a negative z value.

what Does a standard score tell us?

- A standard score or z score of a measurement tells us the number of standard deviations the measurement is from the mean.
  - A standard score close to zero tells us the measurement is near the mean of the distribution.
  - A positive standard score tells us the measurement is above the mean.
  - A negative standard score tells us the measurement is below the mean

A pizza parlor franchise specifies that the average (mean) amount of cheese on a large pizza should be 8 ounces and the standard deviation only 0.5 ounce. An inspector picks out a large pizza at random in one of the pizza parlors and finds that it is made with 6.9 ounces of cheese.

Assume that the amount of cheese on a pizza follows a normal distribution. If the amount of cheese is below the mean by more than 3 standard deviations, the parlor will be in danger of losing its franchise.

How many standard deviations from the mean is 6.9? Is the pizza parlor in danger of losing its franchise?

$$z = \frac{x - \mu}{\sigma} = z = \frac{6.9 - 8}{0.5} = -2.20 \tag{5}$$

Interpretation: The amount of cheese on the selected pizza is only 2.20 standard deviations below the mean. The fact that z is negative indicates that the amount of cheese is 2.20 standard deviations below the mean. The parlor will not lose its franchise based on this sample.

- Given an x distribution with mean m and standard deviation s, the raw score x corresponding to a z score is

$$x = z\sigma + \mu \tag{6}$$

Example

Rod figures that it takes an average (mean) of 17 minutes with a standard deviation of 3 minutes to drive from home, park the car, and walk to an early morning class.

a. One day it took Rod 21 minutes to get to class. How many standard deviations from the average is that? Is the z value positive or negative? Explain why it should be either positive or negative.

b. What commuting time corresponds to a standard score of $z = -2.5$? Interpretation Could Rod count on making it to class in this amount of time or less?

Solution

a.

$$z = \frac{21 - 17}{3} \approx 1.33$$

The z value is positive. We should expect a positive z value, since 21 minutes is more than the mean of 17.

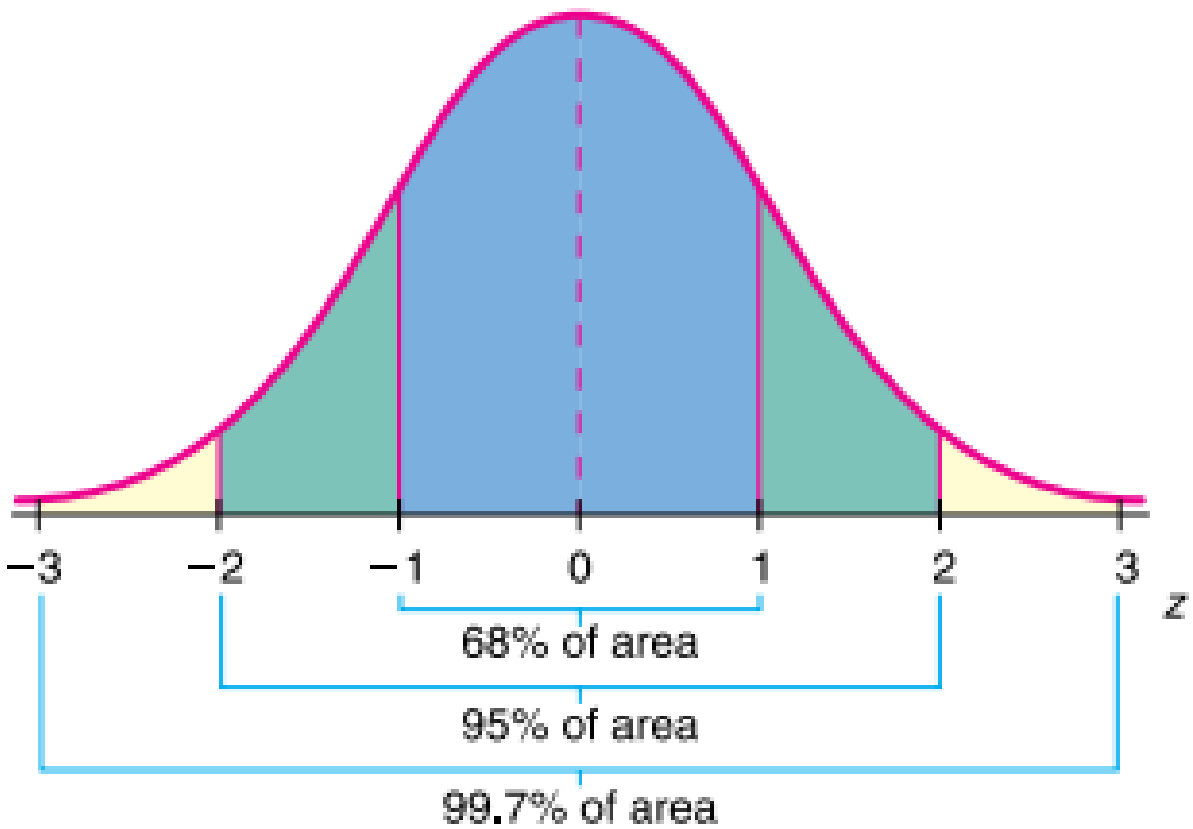
b.

$$x = (-2.5)(3) + 17 = 9.5 minutes$$

No. Commute times at or less than 2.5 standard deviations below the mean are rare.

Standard Normal Distribution

- If the original distribution of x values is normal, then the corresponding z values have a normal distribution as well. The z distribution has a mean of 0 and a standard deviation of 1.



- Find the area between $z = 1.00$ and $z = 2.70$.
  (Area between 1.00 and 2.702) =
  (*Area left of 2.702*) − (*Area left of 1.00*)
  From the Standard normal table;

$$= 0.9965 - 0.8413$$

$$= 0.1552$$

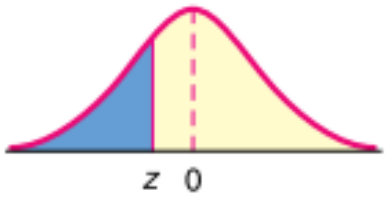
- Find the area to the right of z = 0.94.
  (Area to the right of 0.94) =
  (*Area under entire curve*) − (*Area left of 0.94*)
  From the Standard normal table;

$$= 1.0000 - 0.8264$$

$$= 0.1736$$

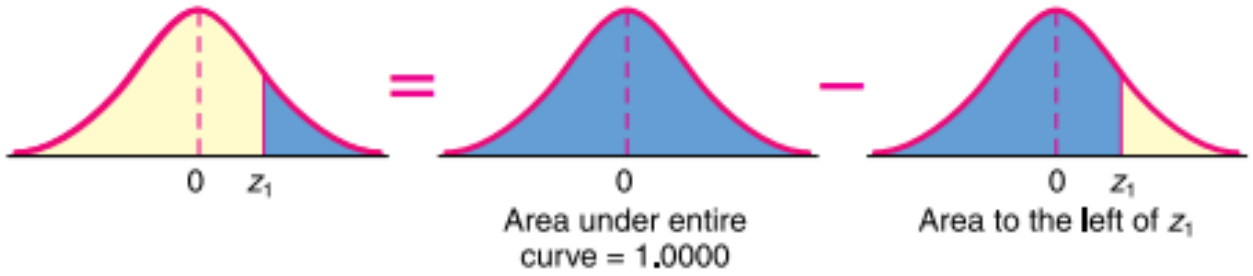
Summary

(a) Area to the left of a given z value



Use Table 5 of Appendix II directly.

(b) Area to the right of a given z value



(c) An alternative approach is to use symmetry of the normal distribution.



(d) Area between two z values



Normal Distribution Probability

The life span of a rechargeable battery is the time before the battery must be replaced because it no longer holds a charge. One tablet computer model has a battery with a life span that is normally distributed with a mean of 2.3 years and a standard deviation of 0.4 year. What is the percentage of batteries to be replaced during the guarantee period of 2 years?

- Let x represent the battery life span. The statement that the battery needs to be replaced during the 2-year guarantee period means the life span is less than 2 years, or $x \leq 2$. Convert this statement to a statement about z.

$$z = \frac{x - \mu}{\sigma} = \frac{2 - 2.3}{0.4} = -0.75 \tag{7}$$

- Hence, $p(z \leq -0.5) = 0.2266$ meaning the company will repair or replace about 23% of their batteries

THANK YOU!!!